NAME: _____ Geometry (FL B.E.S.T.)

1.3 Measuring and Constructing Angles

Angle measure turns a picture of two rays into a number you can compute with, the same way segment length did for two points. The Angle Addition Postulate and the definition of an angle bisector let you build equations with a variable buried inside a measure, not just read a protractor. This section also connects back to earlier ideas — opposite rays make a straight angle, and a bisector's two halves are congruent — so several problems ask you to justify a step, not only find a number.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Classify $m\angle A = 72°$.

2. Classify $m\angle B = 90°$.

3. Classify $m\angle C = 140°$.

4. Measure of a straight angle?

5. $m\angle ABD = 40°$, $m\angle DBC = 55°$. Find $m\angle ABC$.

6. $m\angle ABC = 100°$, $m\angle ABD = 65°$. Find $m\angle DBC$.

7. $\overrightarrow{BD}$ bisects $\angle ABC$, $m\angle ABD = 32°$. Find $m\angle ABC$.

8. $m\angle ABD = 4x$, $m\angle DBC = 2x + 18$, bisector. Find x.

9. $m\angle ABD = x$, $m\angle DBC = 2x$, $m\angle ABC = 84°$. Find x.

10. Classify $m\angle D = 180°$.

11. $m\angle PQR = 130°$, $m\angle PQS = 80°$. Find $m\angle SQR$.

12. $m\angle ABC = 70°$, $\overrightarrow{BD}$ bisects. Find $m\angle ABD$.

13. $m\angle ABD = 5x - 4$, $m\angle DBC = 3x + 8$. Find x.

14. Ray BA reads $15°$ and ray BC reads $162°$ on the same protractor scale. Find $m\angle ABC$ and classify it.

15. $\overrightarrow{BD}$ bisects $\angle ABC$, and $\overrightarrow{BE}$ bisects $\angle ABD$. If $m\angle EBD = 18°$, find $m\angle ABC$.

✓ Answer Key

1. Acute
2. Right
3. Obtuse
4. $180°$
5. $95°$
6. $35°$
7. $64°$
8. $x = 9$
9. $x = 28$
10. Straight
11. $50°$
12. $35°$
13. $x = 6$
14. $147°$, obtuse
15. $72°$